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Generative AI interactive textbook in electrotechnics: A four-year comparative study on student performance and inclusion

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ABSTRACT

This study presents the results of the implementation of Generative-AI Interactive Textbook, an intelligent textbook built on a large language model based on GPT-4, which is directly integrated into the subject of Electrical Engineering course and provides interactive, adaptive teaching functions. The textbook was deployed in the 2024/2025 academic year, and the data was compared with the summary results from 2021 to 2024. The sample consisted of 736 students.

The results suggest that the effects associated with the introduction of GenAI-T vary depending on the type of assessment and the analytical context. In multi-year analyses, a statistically significant improvement in mid-term assessment results was consistently observed, both across the overall student population and within individual subgroups. In contrast, final assessment results showed a more variable pattern, with a statistically significant difference identified in a one-year comparison between groups with and without access to GenAI-T, while no such association was observed in multi-year analyses.

In this study we present a practical method for implementing the Intelligent Textbook, which includes analyzing course objectives, creating targeted prompts for configuring the AI assistant, and linking it to existing forms of education, thereby ensuring smooth integration into teaching and a reliable framework for further testing of the system. At the same time, it sets out principles to ensure that generative AI truly develops cognitive abilities, independence, and critical thinking, including the didactic anchoring of tasks to higher levels of Bloom's taxonomy, active verification and reflection on AI outputs, preservation of student authorship, and transparency in the use of technology.

The results indicate that a systematically designed generative AI Interactive Textbook is associated with improved outcomes during the semester, with the most consistent increase observed in mid-term assessment, while effects on final assessment and between-group differences remain indicative rather than conclusive.

1. Introduction

OpenAI release of ChatGPT in November 2022 represented a significant shift in the social acceptance of large language models (LLM). Although ChatGPT is often considered the most representative product of generative artificial intelligence (GenAI) (Ma, 2025), various large language models now play important roles in natural language processing with varying strengths in accuracy and output naturalness (Janakiraman & Ghoraani, 2025). These models have rapidly been adopted across various domains, including higher education (Ma, 2025).

One of the key benefits is personalized and adaptive learning, where content is tailored to students' individual needs, preferences, and pace (Chun et al., 2025). LLMs can generate questions that stimulate students' interest and provide immediate, contextually tailored responses, simulating a private tutor (Chan & Hu, 2023). AI-supported systems such as adaptive learning platforms (ALP) also take advantage of this, dynamically adapting content delivery and assessment methods based on continuous analysis of student data (Francis et al., 2024).

The use of LLM and GenAI supports higher levels of student engagement and motivation through interactive and dynamic learning environments (Chun et al., 2025). They offer a safe environment for practicing skills, which can build confidence and fluency in learning. AI tools have been shown to increase student engagement at behavioral, cognitive, and emotional levels (Heung & Chiu, 2025).

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Another benefit is the streamlining of teaching activities using AI tools to automate routine tasks (Francis et al., 2024), tailor teaching content to different types of students, and improve feedback (Bagherimajd & Khajedad, 2025). These technologies allow educators to devote more time to individual student support and the creation of innovative teaching strategies.

In addition to personalization, motivation, and support for teaching activities, AI tools also contribute to the development of basic academic and cognitive skills. They enable students to improve their language skills in a targeted manner, including stylistics, grammar, translation, and the formulation of ideas in a foreign language (Heung & Chiu, 2025). At the same time, if students are guided to reflect on and verify the generated outputs, these technologies can significantly strengthen critical thinking and problem-solving skills (Francis et al., 2024). Finally, generative AI tools can foster creativity by suggesting topics for creative writing or generating multimedia content, thereby broadening the repertoire of instructional strategies beyond traditional paradigms (Francis et al., 2024).

Growing demand for AI skills in professional contexts indicates that college graduates will need AI literacy for employment (Linkedin, 2025). At the same time, a 2024 national survey in the US showed that nearly 27% of workers were already using generative AI directly in their work (Galeano et al., 2025) pointing to the rapid adoption of these technologies across various sectors.

AI tools have been associated with increased workplace productivity (Dell'Acqua et al., 2023). Up to 76% of global companies that have adopted generative AI have seen significant time savings, allowing them to focus on innovation and higher-value work (Linkedin, 2025).

Generative AI has significantly impacted software development (Daniotti et al., 2025), with developers using AI primarily for repetitive tasks and discovering new software libraries, contributing to higher satisfaction and more efficient adoption of new technologies (Li et al., 2024).

These findings suggest that AI is becoming a key factor shaping the future of work. Generative AI has the potential to significantly transform work processes (Mayer et al., 2025). This highlights the importance not only of implementing these technologies in the workplace, but also of integrating them into higher education, which prepares students for successful employment.

2. Problem statement and study motivation

Despite the extensive benefits and growing importance of large language models in education, significant gaps remain in the research and implementation of these technologies, particularly in higher education (Chun et al., 2025). Existing solutions often use AI only as a supplementary feature, with some studies focusing on AI-driven textbooks in technical subjects showing no statistically significant differences in academic outcomes. There is a lack of thorough research on how these tools can proactively develop students' complex digital competencies (Qian, 2025), including critical thinking and interaction with AI, as overreliance may hinder independent learning (Consoli & Petko, 2025). There is still a lack of comprehensive empirical data on how the use of generative AI affects internal differences in academic performance among different groups of students. Concerns also remain regarding the potential accumulation of cognitive debt, where overreliance on AI may undermine active critical thinking and lead to passive consumption of information, suggesting a need for more robust research in this area (Qian, 2025). Generative AI is a promising language support tool for students studying in a foreign language. There is an indication that it can help explain concepts, translate, and formulate academic texts, which can mitigate language and cultural barriers (Qian, 2025). Nevertheless, there is a lack of systematic research to quantitatively verify its actual benefits for students with limited language proficiency.

In an effort to effectively address these challenges, the scientific community is gradually turning its attention to more advanced and systematic implementations of artificial intelligence in educational tools. One promising way to overcome these gaps is the concept of so-called intelligent textbooks. Recent developments in this area have shown that textbooks have undergone significant changes, from the first generation based on adaptive systems, through integration with external sources and automation of content creation, to the latest generation of textbooks that are based on machine learning algorithms. While earlier generations focused mainly on predefined adaptivity and content structuring, they offered limited support for open-ended dialogue, flexible explanations, and real-time cognitive scaffolding, which are essential in complex STEM (science, technology, engineering and mathematics) learning contexts. This latest generation of intelligent textbooks uses large language models to generate interactive content, personalized feedback, and simulate dialogues between students and AI assistants (Sosnovsky et al., 2025). Recent empirical studies report that intelligent texts enhanced with large language models can support increased interactivity and measurable learning gains, providing early evidence for the educational potential of generative approaches beyond traditional intelligent textbook paradigms (Crossley et al., 2025). However, the concrete practical implementation of these advanced concepts in STEM subjects, as well as a thorough evaluation of their effectiveness, still represents a challenge and a significant research gap.

To address current challenges in the field of intelligent textbooks, we implemented a practical approach based on large language models, resulting in an interactive assistant named Generative Artificial Intelligence Textbook (GenAI-T). The necessity of introducing generative AI in this context lies in its ability to provide dialogic interaction, adaptive explanations, and scalable personalization that earlier intelligent textbook generations could not sufficiently support. This system builds upon the principles of fifth-generation intelligent textbooks and supports personalized and adaptive learning while its use also enhances students' digital competencies.

In our work, we focus on three research questions that address selected aspects of the identified research gaps:

- **RQ1:** How did the results in the mid-term and final assessments of the Electrical Engineering course change after the introduction of GenAI-T compared to previous academic years?
- **RQ2:** What differences in academic performance were observed in the 2024/2025 academic year between student groups with official access to GenAI-T and groups without such access?
- **RQ3:** How have the differences in academic performance between domestic and foreign students changed since the introduction of GenAI-T compared to previous academic years?

In this study, we will also present an effective way to implement GenAI-T into the teaching process using GPTs technology from OpenAI.

2.1. Pedagogical risks and considerations of generative AI

Before describing the GenAI-T system and its implementation in the teaching process in more detail, it is necessary to pay attention to the potential negative aspects of large language models. A thorough understanding of these aspects enables a responsible approach to their use and, at the same time, creates space for identifying and eliminating risks that could arise from their inappropriate or uncritical application.

The uncritical use of LLM and generative AI in education may pose several risks that could potentially negatively affect students' academic performance and cognitive development. Preliminary findings suggest that one of the main risks is a reduction in cognitive activity. For example, experimental electroencephalography (EEG) measurements have indicated that students using GenAI tools to write essays showed lower brain activity, particularly in networks related to working memory and executive functions. Studies also suggest that over time, students may become more superficially engaged in tasks, more often adopting

AI-generated outputs rather than engaging in their own creative interaction. This often results in homogeneous work that lacks originality and personal input, as confirmed by teachers who perceived these works as less authentic (Kosmyna et al., n.d.).

Another related issue is weaker comprehension and memory, as students who relied on AI had trouble explaining their own texts and showed signs of superficial learning, likely due to insufficiently deep processing of information (Kosmyna et al., n.d.). A study with high school students in mathematics, a key STEM subject, showed that although the AI tutor improved their performance during exercises (short-term productivity), they achieved significantly worse results in a subsequent test without AI assistance, indicating a disrupted knowledge acquisition process and reduced skill acquisition. (Bastani et al., n.d.). There is also a persistent risk of automation bias (overreliance), where excessive trust in AI leads to a reduction in critical thinking and cognitive effort. (Kosmyna et al., n.d.). Students often fail to recognize errors or inaccuracies generated by AI, nor do they consistently verify the information for accuracy. Large language models (LLMs) can produce false, outdated, or misleading content commonly referred to as 'hallucinations', which may lead to misconceptions if accepted uncritically (Wu et al., 2025). This lack of AI literacy is considered a significant challenge (Qian, 2025).

Ultimately, overreliance on AI may diminish student motivation and engagement, leading to an illusion of competence that is, a false sense of mastery that only becomes evident when learners are assessed independently (Qian, 2025). Long-term and uncritical use of AI may even weaken the neural pathways in the brain responsible for memory, problem solving, creativity, and critical thinking, posing a threat to the development of basic cognitive abilities (Kosmyna et al., n.d.). It is therefore essential that the integration of AI into education includes clear pedagogical strategies and an emphasis on the development of AI literacy and critical thinking to prevent this cognitive debt.

The negative impact of these technologies can be mitigated through the proper use of LLM and GenAI. Guided use opens up significant opportunities and brings clear benefits for students' cognitive development, thereby mitigating the risks associated with uncritical reliance on AI (Guo & Lee, 2023). Rather than functioning as a mere 'solution machine,' AI should be positioned as a collaborative partner that prompts students to engage in critical thinking and verify outputs rigorously. Educators play a pivotal role in establishing the expectation that AI tools can produce errors or 'hallucinations,' thereby fostering a responsible and reflective approach to AI-assisted learning (Qian, 2025). This approach is essential for overcoming passive consumption of information and the illusion of competence that can arise from overreliance on AI (Guo & Lee, 2023).

It appears that appropriately guided use of AI in education can promote active and independent thinking. Students who worked with AI as a support tool for asking questions, proposing solutions, and evaluating alternatives developed a higher level of critical, reflective, and creative thinking than those who did not use AI. This approach also contributed to a deeper understanding of the subject matter (Essel et al., 2024). AI thus acts as a catalyst for higher cognitive processes, as it requires students to critically assess, analyze, and evaluate artificial intelligence designs, which pushes them to higher levels of thinking, for example, according to Bloom's taxonomy, including analysis, evaluation and creation (Wang & Fan, 2025).

Although there are concerns that AI stifles creativity, especially when used uncritically, when properly guided, AI becomes an important brainstorming partner that can inspire students and encourage their own originality and creativity (Wu et al., 2025). A meta-analysis of 51 experimental and quasi-experimental studies from 2022 to 2025 confirms an overall positive impact on performance and the development of higher cognitive abilities when AI is integrated systematically. This extensive analysis confirmed an increase in students' analytical and creative thinking, emphasizing that the key is the pedagogical framework and targeted tasks (Wang & Fan, 2025). Tasks should be structured to focus on higher cognitive goals, not just memorization, for example, requiring students to explain the logic of the AI solution (analysis), evaluate the correctness of the output (evaluation), or propose an alternative solution (creation). This approach creates a balanced relationship between the student, teacher, and AI, with the teacher's role being to guide the interaction so that students develop critical AI literacy while avoiding overreliance on automated systems (Qian, 2025).

The use of generative AI in education brings both advantages and risks, with the resulting impact on individuals depending on how it is used. Higher education institutions play a key role in this process, as educators should actively shape the environment in which AI contributes to the development of cognitive abilities, independence, and critical thinking. Clearly defined teaching strategies can ensure that positive effects prevail not only at the individual level, but also in the broader social context.

3. Generative AI interactive textbook

In this study, GenAI-T is presented not as a new generative model, but as a pedagogically designed implementation of a large language model–based assistant, integrated with domain-specific materials to support intelligent textbook use in STEM education. Generative Artificial Intelligence Textbook represents a new generation of educational tools that combine the capabilities of large language models (LLMs) with the didactic principles of interactive learning. Unlike previous forms of intelligent textbooks, which relied on predefined answers, adaptive algorithms, or structured knowledge databases, GenAI-T uses the generative capabilities of AI to create dynamic content in real time. This means that students are not just passive consumers of information, but active participants in a dialogue where every question, explanation, or mistake can lead to a new, contextually adapted response.

GenAI-T can serve as:

- A virtual tutor that simulates explanations, questions, and feedback.
- A discussion partner that supports argumentation, comparison of concepts, and critical thinking.
- Task for example generator that enables personalized training according to the student's level.
- Interactive corrector or solution guide that guides the student through multiple steps without revealing the final result.

Through these interaction approaches, GenAI-T promotes guided, dialogic engagement rather than direct answer provision, thereby designed to mitigate commonly reported challenges in students' use of generative AI, such as passive answer copying and superficial conceptual understanding. From a pedagogical perspective, GenAI-T extends traditional textbooks by supporting personalized learning and immediate formative feedback, which are particularly valuable in large-scale higher education settings. In addition, it fosters AI literacy and digital competencies, as students are required to actively engage with, verify, and interpret AI-generated outputs.

GenAI-T was developed using OpenAI's GPTs feature, allowing the creation of a custom GPT-4 based assistant without coding. Configured through a system prompt, the assistant acts as an interactive textbook, responding strictly based on uploaded PDF materials. It communicates in a professional, academic tone and supports learning through sample calculations, visual explanations, and textbook-based diagrams.

System Prompt (GenAI-T AI assistant configuration):

"Your task is to act as an intelligent and interactive textbook for students of technical disciplines. You always respond based solely on the content of PDF documents, which serve as a source of knowledge.

Communicate objectively, clearly, and professionally in the style of an experienced university lecturer. If appropriate, provide step-by-step sample solutions to calculation examples. When explaining, refer to relevant images and diagrams from the textbook to aid the student's understanding."

The result is an interactive AI textbook that supports meaningful dialogue, feedback, and active learning. All it took was simple configuration with a system prompt and PDF materials, specifically electrical engineering lecture notes and exercise instructions. This approach has proven to be scalable and suitable for various subjects and levels of study. The chosen minimalist prompt engineering enabled rapid deployment of the system without technical intervention, while remaining open to further development. Modifying the prompt or materials changes the behavior of the AI, making it an effective tool for adapting teaching strategies.

Equally important is preparing students for responsible use of the system. We established clear guidelines for working with the AI assistant, including its limitations and the need for critical verification of outputs, both essential for the effective integration of LLMs into education. This combination of simple technical setup and pedagogically grounded guidance offers a practical and replicable model for implementing generative AI in higher education.

3.1. Practical procedure for implementing GenAI-T into the higher education process

The process of implementing the GenAI-T system into teaching took place in several interrelated steps, which ensured its pedagogical effectiveness, practical feasibility, and objective evaluation of its impact.

1. Analysis of the content and objectives of the subject. At the beginning, the structure of the curriculum, key concepts, and learning outcomes were defined, in which the AI assistant can meaningfully support student learning.
2. Creation and prompting of custom GPTs. Based on the content and requirements, a custom model was created with a suitably designed prompt that determined the tone, scope, and type of responses in accordance with the didactic intent.
3. Teacher training. Teachers underwent training focused on the possibilities and limitations of the system, how to use it, and how to work with the outputs in evaluation and feedback.
4. Explanation of principles and interaction during the semester. Students were familiarized with the principles of responsible use of GenAI-T, including warnings about possible errors in responses. During the semester, they used the tool to solve assignments and during consultations.
5. Evaluation of final exam results. Comparing study results made it possible to objectively assess whether and how the use of GenAI-T affected performance and understanding of the subject matter.

This approach ensured not only the smooth integration of AI into teaching, but also a reliable framework for further evaluation and optimization of the system.

3.2. Usage methodology of the GenAI-T system

To ensure transparency of the study design, the following subsection describes how the GenAI-T system was used by students during the semester. The GenAI-T system was made available to students throughout the semester as a voluntary learning support tool. Within the scope of this study, it was used during self-study activities, preparation for mid-term assessments, and completion of practice-oriented tasks. The system was not accessible during formal examinations and did not directly affect grading or assessment procedures.

Student interaction with GenAI-T followed a guided, dialogic usage approach defined for this study. Rather than providing final answers, the system was used to support step-by-step explanations, conceptual clarification, and decomposition of problem-solving processes. This interaction mode required students to engage with intermediate reasoning steps and formulate follow-up questions during their learning activities.

To minimize the risk of answer copying and superficial learning, specific pedagogical constraints were applied during the study. GenAI-T was configured to prioritize explanatory feedback and conceptual guidance over the generation of final numerical results or complete task solutions. These constraints were intended to support independent problem solving while preserving academic integrity. Objective usage metrics (e.g., interaction logs or frequency of use) were not systematically collected, however, access to GenAI-T was ensured and its use was actively integrated into classroom activities.

4. Methods

Intelligent textbook based on generative artificial intelligence (GenAI-T) was integrated into the Electrical Engineering course as a supplementary instructional tool. Its implementation aimed to empirically examine potential benefits related to supporting comprehension of technical content and improving accessibility for an increasingly international student population. The Electrical Engineering course focuses on the analysis of electrical circuits, including direct current and alternating current systems, non-sinusoidal waveforms, and transient phenomena, with an emphasis on fundamental principles and analytical methods.

4.1. Methods of the four-year study with all students

The aim of this study was to quantitatively assess student achievement in the Electrical Engineering course across four academic years (2021/2022–2024/2025) and to examine changes associated with the introduction of an AI-based instructional tool (GenAI-T). Student performance was evaluated using two assessment types: mid-term assessment conducted during the semester (40% of the total grade) and a final examination at the end of the course (60%). The study period was segmented into two phases:

- **2021/2022 to 2023/2024:** Instruction was delivered using conventional methods with no implementation of AI-based tools.
- **2024/2025:** GenAI-T—an intelligent textbook optimized for mid-term assessment—was introduced.

Differences in student success rates across research phases were evaluated using Pearson's chi-square test of independence to determine their statistical significance. Given the retrospective and observational nature of the study design, the results should be interpreted as associations rather than causal effects and may be influenced by unobserved confounding factors.

4.2. Methods of the one-year study with experimental and control group

This study examines the impact of GenAI-T on the academic performance of students enrolled in the Electrical Engineering course during the 2024/2025 academic year. The cohort was divided as follows:

- **Experimental group:** Students in groups D4, D6, D7, and D8, who were granted access to the GenAI-T tool.
- **Control group:** Students in groups D1, D2, D3, and D5, who were not granted official access to GenAI-T.

Due to the open nature of the tool and informal information sharing among students, access to GenAI-T could not be strictly controlled. The distinction between experimental and control groups therefore reflects formal access rather than exclusive usage.

Student performance was evaluated using mid-term and final assessments, with success defined as meeting minimum performance thresholds. Pearson's chi-square test of independence was applied to assess the statistical association between access to GenAI-T and student success in both assessment components.

4.3. Methods of the four-year study with domestic and foreign students

This study aimed to compare the academic performance of domestic and foreign students in the Electrical Engineering course across four consecutive academic years, considering changes in cohort composition and the introduction of AI-based instructional support. The analysis was conducted as a retrospective study including all students enrolled in the course between 2021/2022 and 2024/2025 Two groups were analyzed:

- **Domestic students.**
- **Foreign students.**

Student performance in mid-term and final assessments was examined separately for each academic year. In the 2024/2025 academic year, a generative AI–based smart textbook was introduced to support semester assessment. For each group and period, student counts, numbers of successful assessments, and success rates were analyzed, and differences between domestic and foreign students were evaluated using Pearson's chi-square test of independence.

5. Results

The results for the Electrical Engineering course were extracted from the university information system and analyzed in accordance with the methodology described above.

5.1. RQ1: Results of the four-year study with all students

Table 1 presents a summary of the outcomes from mid-term assessment over four academic years. The table includes the total number of enrolled students, the number of students who fulfilled the requirements for mid-term assessment, and the corresponding success rates for each period. In Phase 1 (2021/2022), the lowest enrollment was recorded (137 students), with a mid-term assessment success rate of 61.31%. During Phase 2 (2022/2023 and 2023/2024), the number of enrolled students increased to 214 and 187, respectively, while the success rates for mid-term assessment declined to 55.61% and 57.22%. In Phase 3 (2024/2025), following the implementation of GenAI-T, the number of enrolled students was 198 and the success rate for mid-term assessment reached its highest value, 77.78%.

Table 2 summarizes the results of the final (summative) course assessment across the four academic years. In Phase 1 (2021/2022), the success rate for the final assessment was 59.12%. During Phase 2 (2022/2023 and 2023/2024), the success rates were 50.00% and 51.87%, respectively. In Phase 3 (2024/2025), following the introduction of GenAI-T, the success rate for the final assessment was 56.57%.

The results are visualized in Fig. 1. A comparison of outcomes across the research phases indicates that the introduction of GenAI-T was associated with a substantial improvement in students' success rates in mid-term assessments and a modest increase in success rates in final assessments.

Pearson's chi-square test revealed a statistically significant association between the implementation of GenAI-T and student success in mid-term assessments, $\chi^2(1, N = 736) = 24.38$, $p < 0.001$. In contrast, no statistically significant association was found for final course assessments, $\chi^2(1, N = 736) = 0.61$, $p = 0.433$. These results indicate that the introduction of GenAI-T was associated with higher success rates in mid-term assessment but not in summative assessment outcomes.

Table 1
Student success in mid-term assessment.

Academic year	Number of students	Number of successful students	Percentage of successful students
2021/2022	137	84	61.31%
2022/2023	214	119	55.61%
2023/2024	187	107	57.22%
2024/2025	198	154	77.78%

Table 2
Student success in final assessment.

Academic year	Number of students	Number of successful students	Percentage of successful students
2021/2022	137	81	59.12%
2022/2023	214	107	50.00%
2023/2024	187	97	51.87%
2024/2025	198	112	56.57%

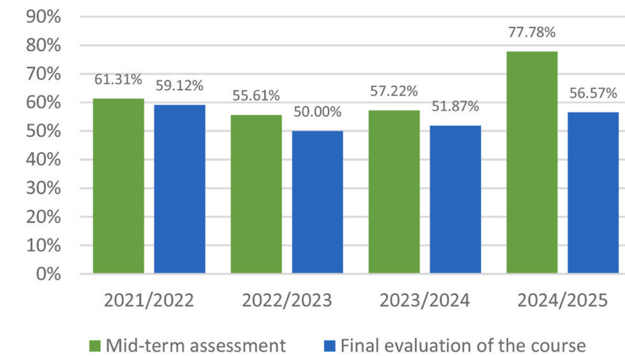

Fig. 1. Visualization of mid-term assessment results and final assessment outcomes across the study period.

5.2. RQ2: Results of the one-year study with experimental and control group

The mid-term assessment results by individual study groups, as presented in Table 3, indicate that the highest success rates were achieved in groups D4 and D5 (100%), while the lowest was observed in group D1 (62.5%).

The summary results presented in Table 4 show that the experimental group (D4, D6, D7, D8) comprised 95 students, of whom 82 met the mid-term assessment criteria, corresponding to a success rate of 86.32%. The control group (D1, D2, D3, D5) consisted of 91 students, with 72 achieving successful completion of the mid-term assessment, representing a success rate of 79.12%.

The results of the final assessment by study group are presented in Table 5. The highest success rates in the final assessment were observed in groups D8 (73.91%) and D5 (72.73%), while the lowest success rate was recorded in group D1 (37.50%).

Table 6 presents the summary results of the final assessment for the experimental and control groups. In the experimental group (D4, D6,

Table 3
Mid-term assessment of study groups.

Group	Study group	Number of students	Number of successful students	Percentage of successful students
Experimental group	D4	23	23	100.00%
	D6	24	23	95.83%
	D7	25	17	68.00%
	D8	23	19	82.61%
Control group	D1	24	15	62.50%
	D2	24	17	70.83%
	D3	21	18	85.71%
	D5	22	22	100.00%

Table 4
Mid-term assessment of experimental and control group.

Group	Number of students	Number of successful students	Percentage of successful students
Experimental group	95	82	86.32%
Control group	91	72	79.12%

Table 5
Student success in final assessment of study groups.

Group	Study group	Number of students	Number of successful students	Percentage of successful students
Experimental group	D4	23	16	69.57%
	D6	24	15	62.50%
	D7	25	15	60.00%
	D8	23	17	73.91%
Control group	D1	24	9	37.50%
	D2	24	10	41.67%
	D3	21	12	57.14%
	D5	22	16	72.73%

Table 6
Student success in final assessment of experimental and control group.

Group	Number of students	Number of successful students	Percentage of successful students
Experimental group	95	63	66.32%
Control group	91	47	51.65%

D7, D8), 95 students were enrolled, with 63 successfully completing the final assessment, corresponding to a success rate of 66.32%. In the control group (D1, D2, D3, D5), there were 91 students, of whom 47 successfully completed the final assessment, representing a success rate of 51.65%.

To enhance clarity and facilitate interpretation, the results of the mid-term assessment are visualized in Fig. 2, while the results of the final assessment are depicted in Fig. 3.

The primary outcome of this part of the study is the comparison of results between the experimental and control groups. These results are visualized in Fig. 4.

Pearson's chi-square test did not reveal a statistically significant difference between experimental and control groups in mid-term assessment outcomes, $\chi^2(1, N = 186) = 1.63$, $p = 0.201$. However, a statistically significant association was observed for final assessment results, $\chi^2(1, N = 186) = 4.14$, $p = 0.042$, indicating higher success rates in the experimental group, though interpretation is subject to the limitations outlined in Section 6.5.

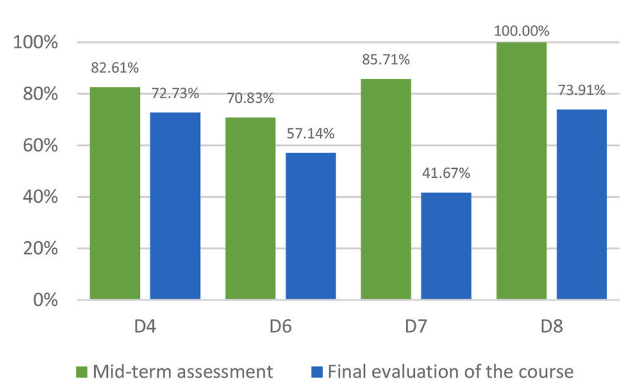

Fig. 2. Visualization of the results of the experimental group.

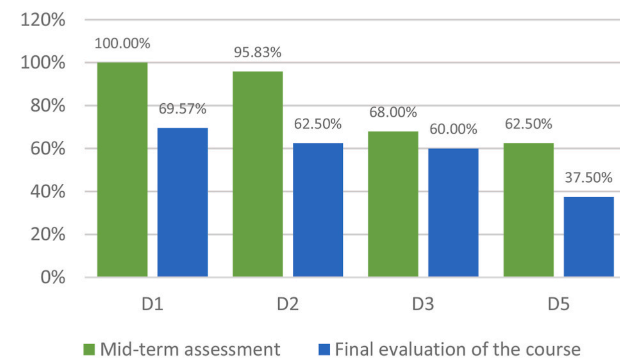

Fig. 3. Visualization of the results of the control group.

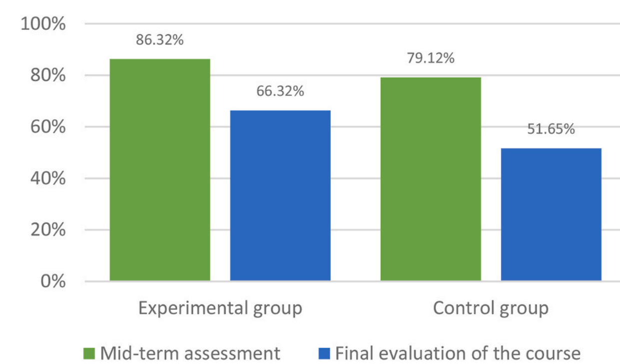

Fig. 4. Comparison of results between the experimental and control groups.

5.3. RQ3: Results of the four-year study with domestic and foreign students

This section presents the quantitative results of the study, including data from both domestic students and international (foreign) students. The results of the mid-term assessment for domestic students are summarized in Table 7. The number of enrolled domestic students ranged from 123 to 157 across the four academic years. The percentage of students successfully completing the mid-term assessment was similar in the first three years (61.72%, 60.98%, and 62.10%), while in 2024/2025 this rate increased markedly to 81.53%.

Table 8 summarizes the results of the final assessment for domestic students. The success rate ranged from 57.72% to 60.48% over the study period, with no significant year-on-year fluctuations observed.

The results of the mid-term assessment for foreign students are presented in Table 9. The number of international students enrolled in the Electrical Engineering course ranged from 9 to 91 across the academic years. The percentage of students successfully completing the semestral assessments from 2021/2022 to 2023/2024 ranged from 46.15% to 55.56%. In the 2024/2025 academic year, the pass rate for international students increased modestly to 63.41%.

The results of the final assessment for foreign students are presented in Table 10. In 2021/2022, only 9 foreign students were enrolled, with a success rate of 55.56%. In subsequent years, the number of international students increased to 91 in 2022/2023, 63 in 2023/2024, and 41 in

Table 7
Domestic student success in mid-term assessment.

Academic year	Number of students	Number of successful students	Percentage of successful students
2021/2022	128	79	61.72%
2022/2023	123	75	60.98%
2023/2024	124	77	62.10%
2024/2025	157	128	81.53%

Table 8
Domestic student success in final assessment of the course.

Academic year	Number of students	Number of successful students	Percentage of successful students
2021/2022	128	76	59.38%
2022/2023	123	71	57.72%
2023/2024	124	75	60.48%
2024/2025	157	91	57.96%

Table 9
Foreign student success in mid-term assessment.

Academic year	Number of students	Number of successful students	Percentage of successful students
2021/2022	9	5	55.56%
2022/2023	91	42	46.15%
2023/2024	63	30	47.62%
2024/2025	41	26	63.41%

Table 10
Foreign student success in final assessment of the course.

Academic year	Number of students	Number of successful students	Percentage of successful students
2021/2022	9	5	55.56%
2022/2023	91	35	38.46%
2023/2024	63	22	34.92%
2024/2025	41	21	51.22%

2024/2025. The success rates in 2022/2023 and 2023/2024 were similar, at 38.46% and 34.92%, respectively. In 2024/2025, the success rate increased to 51.22%.

To aid in the interpretation of the data, the results described above are visualized in Figs. 5 and 6.

Pearson's chi-square analyses revealed statistically significant improvements in mid-term assessment outcomes following the implementation of the GenAI-T for both domestic students, $\chi^2(1, N = 532) = 21.67$, $p < 0.001$, and international students, $\chi^2(1, N = 204) = 4.91$, $p = 0.027$. In contrast, no statistically significant associations were observed for final assessment outcomes. For domestic students $\chi^2(1, N = 532) = 0.31$, $p = 0.58$ and for foreign students $\chi^2(1, N = 204) = 3.52$, $p = 0.061$.

6. Discussion

This section presents the main findings of the study, interprets them in relation to the research questions, and assesses their practical implications, limitations, and recommendations for future research. The discussion is organized according to the specific research questions and addresses the effects of changes in student population, the implementation of GenAI-T, and differences between student groups observed

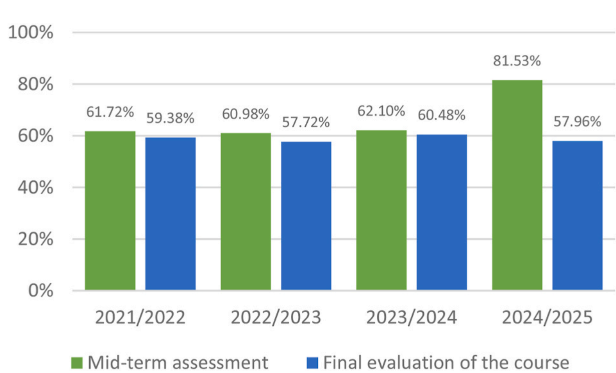

Fig. 5. Visualization of domestic students' results.

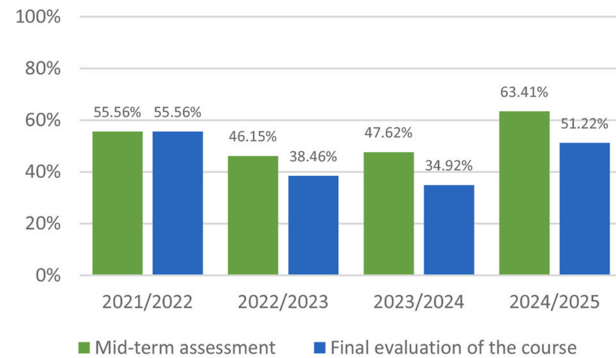

Fig. 6. Visualization of foreign students' results.

during the study period.

6.1. *RQ1: Discussion of the four-year study of all students*

6.1.1. *Principal findings*

The chi-square analysis performed for RQ1 showed a statistically significant correlation between the introduction of GenAI-T and student success in mid-term assessments during the four-year observation period. The distribution of successful and unsuccessful midterm results in the 2024/2025 academic year differed significantly from the results observed in previous academic years, $\chi^2(1, N = 736) = 24.38$, $p < 0.001$. This result suggests that GenAI-T may be associated with higher mid-term success rates, however causal effects cannot be inferred due to the retrospective and observational nature of this study.

Analysis of the final assessment results did not reveal a statistically significant correlation between the assessment period and student performance, $\chi^2(1, N = 736) = 0.61$, $p = 0.433$. The distribution of the final assessment results remained statistically comparable in all academic years examined.

6.1.2. *Practical implications*

From a practical point of view, the results show that statistically significant changes related to the introduction of GenAI-T were observable for the assessed area for which the smart textbook was optimized, while in the same longitudinal comparison, no statistically significant changes were found in the assessment results for the area for which Gen-AI-T was not optimized.

6.2. *RQ2: Discussion of the one-year study with experimental and control group*

6.2.1. *Principal findings*

For RQ2, Pearson's chi-square test did not identify a statistically significant difference in mid-term assessment success between the experimental and control groups within the same academic year, $\chi^2(1, N = 186) = 1.63$, $p = 0.201$. This finding indicates that mid-term assessment outcomes were statistically comparable between groups.

It is noteworthy that the analysis of final assessment outcomes revealed a statistically significant association between experimental group membership and student success, $\chi^2(1, N = 186) = 4.14$, $p = 0.042$. The distribution of successful and unsuccessful final assessment outcomes differed significantly between the experimental and control groups.

6.2.2. *Practical implications*

These results indicate that, within a single academic year, statistically significant differences between students with and without access to GenAI-T emerged only in final assessment outcomes. No statistically supported group-level differences were observed for mid-term assessments. This result indicates that the observed differences may have been influenced by additional factors beyond the use of GenAI-T. Further research in this area will be necessary.

6.3. RQ3: Discussion of the four-year study with domestic and foreign students

6.3.1. Principal findings

The chi-square analyses addressing RQ3 revealed statistically significant changes in mid-term assessment outcomes following the introduction of GenAI-T for both domestic students, $\chi^2(1, N = 532) = 21.67$, $p < 0.001$, and foreign students, $\chi^2(1, N = 204) = 4.91$, $p = 0.027$. In both cases, the distribution of successful and unsuccessful mid-term outcomes differed significantly between the pre-intervention period and the academic year in which GenAI-T was introduced.

In contrast, no statistically significant changes were identified for final assessment outcomes. For domestic students, the association between assessment period and success was not statistically significant, $\chi^2(1, N = 532) = 0.31$, $p = 0.58$. Similarly, for foreign students, the observed difference in final assessment outcomes did not reach statistical significance, $\chi^2(1, N = 204) = 3.52$, $p = 0.061$.

6.3.2. Practical implications

The results indicate that statistically significant changes associated with the introduction of GenAI-T were confined to mid-term assessments for both domestic and foreign students. Final assessment outcomes for both groups remained statistically comparable to those observed in previous academic years.

6.4. Discussion of all studies

The results of RQ1, RQ2, and RQ3 suggest that the observed effects are not consistent across different types of assessment or analytical approaches. While multi-year studies (RQ1 and RQ3) revealed statistically significant associations for mid-term assessments but not for final outcomes, the one-year study (RQ2) showed the opposite pattern.

These differences can be explained by variations in study design, sample size, and the level of data aggregation across the studies. RQ1 and RQ3 are based on aggregated data spanning multiple academic years and cohorts, whereas RQ2 compares groups within a single academic year under identical time conditions. The results thus suggest that the observed associations depend on the analytical framework used and should be interpreted in the context of the respective study design.

6.5. Pedagogical implications and learning theory perspective

Taken together, the results presented in Sections 4.1 – 4.3 provide a basis for discussing the observed performance patterns from a pedagogical point of view. The way students interacted with GenAI-T through guided, step-by-step explanations aligns with instructional scaffolding, which supports learning by breaking complex concepts into manageable parts (Belland et al., 2017). The stronger effects observed in mid-term assessments are consistent with formative feedback approaches, where continuous support during the semester helps students identify and address misunderstandings earlier, rather than relying mainly on support at the end of the course (Hattie & Timperley, 2007).

In addition, the voluntary and dialogic use of GenAI-T reflects elements of self-regulated learning, as students decided when and how to use the system to support their study activities. Together, these perspectives help clarify why GenAI-T appears to be more closely associated with improvements in formative rather than summative assessment outcomes and allow the results to be understood in terms of learning processes rather than technology alone.

6.6. Limitations & future work

Given the retrospective and observational nature of this study, the findings should be interpreted as associations rather than causal effects. When comparing data within a single year, it was not possible to strictly control access to GenAI-T in an open educational environment. As a result, the informal use of this tool among students in the control group cannot be completely ruled out. Consequently, the potential influence of unobserved confounding variables must be considered when interpreting the findings. Changes in the composition of the student population at different stages may have affected the results. Other factors, such as language proficiency and prior educational background, may also have influenced student outcomes.

For future research, we recommend the use of randomized controlled studies with clearly defined conditions for access to AI tools. Longitudinal designs and the integration of both quantitative and qualitative methods are also advisable to better capture the effects of AI on academic performance and its long-term benefits. Further research should address linguistic, cultural, and adaptation factors, and more closely examine the ways in which students utilize AI tools during their studies.

In parallel with methodological advances, future work will focus on further development of the GenAI-T system itself. Planned activities include systematic comparison of different large language models, including both commercial and open-source solutions, with emphasis on output accuracy, pedagogical suitability, and domain relevance, as well as support for local deployment in university environments. Additional efforts will target the analysis of usage metrics, such as interaction frequency and task types, to support more effective personalization and to promote AI literacy through structured guidance on responsible and effective AI use. These steps aim to support scalable and sustainable integration of GenAI-T into higher education.

7. Conclusion

A comprehensive quantitative and comparative analysis of outcomes in the Electrical Engineering course across four academic years indicates that the introduction of GenAI-T may be associated with changes in student performance, particularly in mid-term assessment outcomes. The largest gains were observed during the semester, and the introduction of GenAI-T was accompanied by a descriptive reduction in performance differences between domestic and international students. These findings suggest that personalized, adaptive support may contribute to ongoing learning, particularly for students facing greater academic stress or with specific learning needs.

GenAI-T appeared more effective in areas aligned with its optimized content and feedback, namely mid-term assessment. Effects on the final course assessment were positive but smaller, underscoring the need to extend and adapt AI support to address the full breadth and complexity of end-of-course requirements.

The study provides indicative evidence that AI-based instructional tools may be associated with reduced educational disparities in authentic instructional settings and underscores the importance of further research on how AI tools are used in real-world learning contexts.

Overall, the findings suggest a potential role of generative AI tools in promoting more inclusive, adaptive, and effective university instruction. Realizing this potential will require systematic optimization and scaling of these tools in the context of increasing diversity and digitalization in higher education.

Finally, successful implementation depends on sound pedagogical integration, including clear usage guidelines, student-facing guidance, and methodological support. Such practices not only deepen subject understanding but also cultivate the digital competencies needed for effective interaction with LLM-enabled technologies in contemporary university teaching.

CRediT authorship contribution statement

Branislav Fecko: Writing – original draft, Methodology,

Conceptualization. **Jozef Dziak:** Writing – original draft, Formal analysis, Data curation. **Tibor Vince:** Writing – review & editing, Visualization, Validation. **Ján Molnár:** Writing – review & editing, Supervision, Funding acquisition.

Ethics statement

This study involved fully anonymous educational data. A voluntary online questionnaire was conducted to obtain student feedback on the use of the GenAI-T system. These data were not included in the quantitative analysis presented in this study. In accordance with institutional guidelines, formal ethical approval was not required. Informed consent was obtained from all participants, and their privacy and data protection rights were strictly observed.

Declaration of generative AI and AI-assisted technologies in the writing process

OpenAI's ChatGPT was employed to assist with language refinement and to enhance the clarity of the manuscript. All authors carefully reviewed and revised the text following the use of the AI tool and accept full responsibility for the integrity and accuracy of the published content.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

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